

Canonical-prefix rank, active-spectator lift, and relative-Hessian boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-147 found an exact active-spectator direction on which the retained production $P+L$ coefficient energy is not convex. The next proposed step was to place that direction in an R-146 canonical two-root chart and append the shared trace, Doob bracket, R-063 forest, future variance, balanced rows, returned low, and terminal sextic exactly once. R-148 proves that this proposal was not yet a well-defined computation. There are four conclusions.

1. The fresh final scalar innovation of the exact R-147 line cannot be a nonzero final increment of the R-146 proportional-covariance canonical chart. On the active-spectator plane every nonzero canonical increment has rank two, while that fresh line innovation has rank one. A rank-one past block is only the no-correction line fixture, not a necessity theorem for an adapted past.
2. This is not a no-go for a canonical Brownian chart. It means that a new canonical chart necessarily leaves the exact line and must recompute the full-rank coefficient, forest, low, and spatial owners. Missing companions may not be set to zero or imported from the old chart.
3. A separately declared noncanonical last-root Gaussian coefficient completion is exactly computable. At each fixed point strictly inside the R-147 adverse region, its predictable trace-bracket mismatch is positive for all sufficiently small noise. Thus even the toy lift fails automatic trace-bracket matching.
4. The R-147 negative second derivative is an absolute-owner diagnostic, not a physical deterministic-control Hessian. In the declared coefficient- background parameter family the parameter gradient uses f''' and its Hessian uses $f^{(4)}$. At each fixed adverse point and sufficiently small noise the owner and terminal-sextic curvature are positive and the source curvature is nonnegative. In the orientation $R > 0$ the first derivative is negative; invariantly it is nonzero. The origin is therefore not stationary in this diagnostic, but no physical control Hessian, local minimum, or phase is identified.

The result is phase-neutral. BCC, a different modulated state, a uniform state, or inadequacy of the present effective action are all downstream possibilities. None is preferred or excluded here.

2. Registered coordinates and ownership firewall

Fix one positive floor $e > 0$, oriented base radius $R > 0$, production normalization $P > 0$, and the A1 coefficients

$$\alpha = \frac{5}{9}, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}. \quad (2.1)$$

R-125 defines the unaveraged six-row current, exhaustive output projections, primitive trace, conditional future variance, and the adapted R-063 forest. R-141 then packages the complete trace/current/future

owner in one signed feature. Future variance and forest are alternate coordinates of that owner; they are not extra favourable reserves.

R-142 supplies two different objects which must not be conflated:

- the exact pointwise six-row coefficient and shared-probe trace feature;
- a legal finite-chart assembly recipe whose production cross-Grams, spatial projectors, adapted feed-back, and returned-low blocks are not filled.

R-146 supplies a fresh canonical endpoint-law chart with increments

$$\Delta C_b = \tau_b C_J. \quad (2.2)$$

It preserves the endpoint variational value, not the individual old reveal, trace, forest, variance, or low owners. R-147 supplies a canonical Doob tower for a newly declared terminal feature, but owner transport still requires the tower, incidence, and predictable trace–bracket match.

Accordingly, R-148 uses two separate scopes:

1. an exact rank theorem about the R-146 canonical covariance;
2. a newly declared last-root coefficient diagnostic, not a transported production owner.

3. Canonical-prefix rank obstruction

The registered A1 internal mass block is

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \quad (3.1)$$

Its leading principal minors are

$$\frac{1}{10}, \quad \frac{21}{2000}, \quad \frac{3}{3125}, \quad (3.2)$$

so $M \succ 0$. At scalar kinetic eigenvalue $a \geq 0$ the covariance is

$$C(a) = (aI + M)^{-1} \succ 0. \quad (3.3)$$

Let E be the active–spectator plane spanned by the first and third internal coordinates. The restriction C_E is positive definite. The exact complementary-minor calculation gives

$$\det C_E(a) = \frac{250(100a + 13)}{25000a^3 + 10000a^2 + 1115a + 24} > 0. \quad (3.4)$$

Write

$$w = (1, 1), \quad v = (1, -1). \quad (3.5)$$

The exact R-147 conditional line has the form

$$W = Rw + qv. \quad (3.6)$$

In the no-correction two-root line fixture, the common base and fresh active–spectator fluctuation have covariance supports span w and span v . Their blocks have forms

$$\lambda ww^T, \quad \mu vv^T, \quad (3.7)$$

and each nonzero block has rank one. An adapted past with later correction need not have the first form in (3.7), so no past-rank necessity is asserted. The fresh final scalar innovation, however, necessarily has the second form.

Theorem 3.1 (exact proportional-prefix incompatibility)

Assume $C_E \succ 0$ and the fresh final canonical block has $\tau_2 > 0$. Then

$$\text{rank}(\tau_2 C_E) = 2. \quad (3.8)$$

Consequently it cannot equal the rank-one fresh covariance $\mu v v^T$ in (3.7). The exact R-147 scalar final innovation cannot be inserted unchanged as a nonzero final block of the R-146 proportional-covariance canonical chart.

Proof. Equation (3.4) gives $\det C_E > 0$. Therefore $\det(\tau_2 C_E) = \tau_2^2 \det C_E > 0$, while $\det(\mu v v^T) = 0$. $\square$

The theorem has a strict boundary. It does not say that a canonical Brownian chart fails to exist. In the R-146 chart every nonzero block carries both radial directions. The resulting state is not confined to (3.6), and the full coefficient map must be evaluated again. Adding a complementary Gaussian to restore the missing covariance has exactly the same effect: it moves the state off the R-147 line.

4. A generic last-root mismatch identity

A noncanonical fixed chart can isolate the active-spectator line. The following calculation is a coefficient diagnostic and is not identified with the old R-063/R-125 production owner.

Put

$$x(q) = R + q, \quad y(q) = R - q, \quad d(q) = x(q)^2 + y(q)^2 + e, \quad (4.1)$$

and define the two retained rows

$$A(q) = \left(2\sqrt{c_0} x(q), 2\sqrt{c_1} \left\{ x(q) - \alpha \frac{x(q)^2}{d(q)} [x(q) - y(q)] \right\} \right). \quad (4.2)$$

Then

$$f(q) = \|A(q)\|^2 \quad (4.3)$$

is exactly the R-147 retained $P + L$ energy.

Let g, ζ be independent standard Gaussians, $q = \sigma g$, and define

$$U = \sigma A(q)\zeta, \quad \Phi = \sigma g A(q), \quad b = \mathbb{E}\Phi. \quad (4.4)$$

The shared-probe trace and centered Doob bracket are

$$\bar{\tau} = \mathbb{E}\|U\|^2 = \sigma^2 \mathbb{E}f(\sigma g), \quad (4.5)$$

$$\beta = \mathbb{E}\|\Phi - b\|^2 = \sigma^2 \mathbb{E}[g^2 f(\sigma g)] - \|b\|^2. \quad (4.6)$$

Gaussian integration by parts gives

$$b = \sigma^2 \mathbb{E}A'(\sigma g) \quad (4.7)$$

and therefore

$$\delta := \bar{\tau} - \beta = -\sigma^4 \mathbb{E}f''(\sigma g) + \sigma^4 \|\mathbb{E}A'(\sigma g)\|^2. \quad (4.8)$$

R-147 proves

$$f''(0) = \frac{3(-528R^4 - 88R^2e + 113e^2)}{1000P(2R^2 + e)^2}. \quad (4.9)$$

Let

$$r_2 = -\frac{1}{12} + \frac{5\sqrt{154}}{132} = 0.3867300623 \dots \quad (4.10)$$

If $R^2/e > r_2$, then $f''(0) < 0$. By continuity, $\mathbb{E}f''(\sigma g) < 0$ for sufficiently small σ . The second term in (4.8) is nonnegative, so

$$\delta > 0 \tag{4.11}$$

for all sufficiently small σ , pointwise at each fixed parameter triple (R, e, P) with $R^2/e > r_2$. No one noise threshold uniform up to the boundary $R^2/e \downarrow r_2$ is asserted.

Thus the generic line realization does not repair the missing matching condition $\bar{\tau} = \beta$. This is a failure of automatic owner transfer, not a failure of the complete action.

5. Absolute owner versus coefficient-background parameter

The sign in (4.9) was previously at risk of being read as a negative physical deterministic-control Hessian. That inference is false: the following family moves only the coefficient background while retaining the declared current prefactor σg .

Introduce a coefficient-background parameter m and put

$$q = m + \sigma g. \tag{5.1}$$

The expected complete coefficient owner is

$$\begin{aligned} \mathcal{P}_{\text{comp}}(m) &= \frac{\sigma^2}{2} \mathbb{E}[f(q)(g^2 - 1)] \\ &= \frac{\sigma^4}{2} \mathbb{E}f''(m + \sigma g). \end{aligned} \tag{5.2}$$

Hence, for this declared parameter family only,

$$\mathcal{P}'_{\text{comp}}(0) = \frac{\sigma^4}{2} \mathbb{E}f'''(\sigma g), \quad \mathcal{P}''_{\text{comp}}(0) = \frac{\sigma^4}{2} \mathbb{E}f^{(4)}(\sigma g). \tag{5.3}$$

Direct exact differentiation gives

$$f'''(0) = -\frac{9R(16R^2 + 27e)}{50P(2R^2 + e)^2} < 0, \tag{5.4}$$

and

$$f^{(4)}(0) = \frac{18(112R^4 + 48R^2e - 9e^2)}{25P(2R^2 + e)^3}. \tag{5.5}$$

With $r = R^2/e$, the positive root in (5.5) is

$$r_4 = -\frac{3}{14} + \frac{3\sqrt{11}}{28} = 0.14106694\dots \tag{5.6}$$

The exact comparisons

$$r_2 > \frac{1}{3}, \quad r_4 < \frac{1}{7} \tag{5.7}$$

follow from $\sqrt{154} > 11$ and $\sqrt{11} < 10/3$. Therefore every R-147 adverse point has $f^{(4)}(0) > 0$.

For each fixed adverse (R, e, P) , there is a $\sigma_0(R, e, P) > 0$ such that for $0 < \sigma < \sigma_0$,

$$\mathcal{P}'_{\text{comp}}(0) < 0, \quad \mathcal{P}''_{\text{comp}}(0) > 0. \tag{5.8}$$

The first inequality is essential. Positive curvature does not make $m = 0$ stationary and does not prove a local minimum.

This is not yet the Hessian of a physical source lift. For example, if the same parameter also changed the current prefactor to $\Phi = (m + \sigma g)A(m + \sigma g)$, the uncentred signed owner would acquire the additional terms

$$m\sigma^2 \mathbb{E}f'(m + \sigma g) + \frac{m^2}{2} \mathbb{E}f(m + \sigma g), \quad (5.9)$$

which change both derivatives at the origin. A production control Hessian therefore requires the missing physical synthesis and source lift.

6. Minimal source and sextic completion

This section adds only the source and terminal sextic which are explicitly defined in the R-141 finite-chart action. It does not append future variance, forest, returned low, or balanced rows as new reserves.

Let the source direction have squared chart norm $\kappa \geq 0$. Its relative cost is

$$\frac{9\kappa}{20}m^2. \quad (6.1)$$

For

$$\Psi(q) = (R + q, 0, R - q), \quad (6.2)$$

the terminal sextic is

$$\frac{3}{20} \mathbb{E} \|\Psi(q)\|^6 = \frac{6}{5} \mathbb{E}(R^2 + q^2)^3. \quad (6.3)$$

Subtracting the value at $m = 0$ gives the exact minimal relative polynomial for this coefficient-background parameter family

$$\begin{aligned} \Delta \mathcal{A}_{\min}(m) = & \frac{9\kappa}{20}m^2 + \frac{\sigma^4}{2} \{\mathbb{E}f''(m + \sigma g) - \mathbb{E}f''(\sigma g)\} \\ & + \frac{18}{5}m^2(R^4 + 6R^2\sigma^2 + 15\sigma^4) \\ & + \frac{18}{5}m^4(R^2 + 5\sigma^2) + \frac{6}{5}m^6. \end{aligned} \quad (6.4)$$

Its exact first derivative at the origin is

$$\Delta \mathcal{A}'_{\min}(0) = \frac{\sigma^4}{2} \mathbb{E}f'''(\sigma g), \quad (6.5)$$

and its Hessian is

$$\Delta \mathcal{A}''_{\min}(0) = \frac{9\kappa}{10} + \frac{\sigma^4}{2} \mathbb{E}f^{(4)}(\sigma g) + \frac{36}{5}(R^4 + 6R^2\sigma^2 + 15\sigma^4). \quad (6.6)$$

At each fixed R-147 adverse point, (5.5)–(5.7) and dominated continuity make all three terms in (6.6) nonnegative and the owner and sextic contributions strictly positive for sufficiently small σ ; the source contribution may vanish when $\kappa = 0$. At the same time (6.5) is negative in the chosen $R > 0$ orientation and, invariantly, nonzero. The honest verdict is therefore

$$\boxed{\text{positive minimal curvature, nonstationary origin, no phase verdict.}} \quad (6.7)$$

Future, cross-reveal, balanced, returned-low, spatial, and R-063 forest terms are omitted by definition of this two-row coefficient diagnostic; none is proved zero in the production action. They may neither be appended as independent favourable reserves nor silently deleted. Identifying their one-use contributions requires the missing spatial Wick lift and complete owner incidence.

7. Coefficient diagonal does not identify the owner

Even the complete diagonal function $f(q) = \|A(q)\|^2$ does not determine the mixed Gram or its derivative energy. Put

$$O(q) = \frac{1}{1+q^2} \begin{pmatrix} 1-q^2 & -2q \\ 2q & 1-q^2 \end{pmatrix}, \quad (7.1)$$

let $A_0(q) = (1, 0)$, and set $A_1(q) = O(q)A_0(q)$. Then

$$O(q)^T O(q) = I, \quad \|A_0(q)\|^2 = \|A_1(q)\|^2 = 1. \quad (7.2)$$

Nevertheless

$$\langle A_0(0), A_0(1) \rangle = 1, \quad \langle A_1(0), A_1(1) \rangle = 0, \quad (7.3)$$

and

$$\|A'_0(0)\|^2 = 0, \quad \|A'_1(0)\|^2 = 4. \quad (7.4)$$

Thus diagonal coefficient data do not recover cross-reveal correlations, shared covariance square roots, or Hessian jets.

For an owner-complete production sign, the missing data are at least:

$$\begin{aligned} & \text{time partition and physical synthesis maps;} \\ & W_q, V_q \text{ with all adapted later feedback;} \\ & \Gamma_\ell, \{\Delta\Gamma_k\} \text{ and exhaustive output/spatial projectors;} \\ & \text{root, visit, and owner incidence;} \\ & \text{common heat, shared probe, and future replicas;} \\ & D_j Z, D_j^2 Z \text{ for the adapted R-063 Wick forest;} \\ & \text{balanced spatial atoms, returned low, and the full terminal field.} \end{aligned} \quad (7.5)$$

These are inputs to the assembly recipe, not functions of (4.3).

8. Proof-route and evidence map

Route	Verdict	Exact boundary or successor
Append canonical companions to the exact R-147 line	failed method	The nonzero final canonical block is rank two; the fresh line innovation is rank one.
Construct a new full-rank canonical chart	open	Recompute the full owner off the R-147 line.
Generic last-root line chart	advanced diagnostic	Exact mismatch (4.8) is pointwise positive for sufficiently small noise.
Read $f'' < 0$ as a physical control Hessian	failed interpretation	The coefficient-parameter gradient uses f''' and Hessian uses $f^{(4)}$; the physical source lift is unspecified.
Minimal source/sextic completion	advanced diagnostic	Hessian positive but origin nonstationary at small noise.
Infer full owner from $f(q)$	failed method	Orthogonal gauge witness (7.1)–(7.4) changes mixed Grams and jets.
Select BCC, uniform, or another phase	excluded	No complete-action minimizer comparison has been performed.

The new proof route forks honestly.

1. **Canonical route.** Accept that every root carries the full registered covariance. Build the actual full-rank terminal current and rederive trace, Doob, forest, future, balanced, low, and sextic owners in that same chart.
2. **Old-chart direct route.** Keep the original temporally faithful rank-one/rank-mixed filtration, use the exact R-125 incidence, and prove the owner-complete relative bound without claiming canonical transport.

Either route must test the scalar residual against

$$\eta < \frac{5}{11}, \quad \zeta < \frac{13}{50}, \quad (8.1)$$

with every owner appearing exactly once. If the complete bound fails, the scalar Class-II action should be reassessed without inferring which phase must replace it.

9. Failed-route records and lessons

Three failure records are load bearing.

1. **Exact canonical active–spectator lift.** Terminal covariance equality does not turn the rank-one fresh final line innovation into a nonzero proportional canonical final block. The rank test must precede any companion calculation; no adapted-past rank theorem is inferred.
2. **Jet-to-owner completion.** A pointwise coefficient norm does not identify mixed endpoint Grams, filtration projections, forest jets, future replicas, or low owners.
3. **Absolute defect as relative Hessian.** The Stein identity for the absolute owner contains f'' , while the declared coefficient-background parameter derivative adds two more derivatives. A physical control Hessian still requires a source synthesis, and positive parameter curvature does not prove stationarity.

These records prevent two opposite errors: promoting a partial adverse sign to a physical instability, and deleting the adverse sign merely because a minimal toy Hessian is positive.

10. Devil’s-advocate review

1. **Objection: the active–spectator negative curvature selects a non-BCC phase. UPHELD against that inference.** R-147 concerns one retained coefficient pair, not complete-action minimizers.
2. **Objection: Theorem 3.1 forbids a canonical Brownian chart. DISMISSED.** It forbids only using the exact rank-one fresh R-147 innovation as a nonzero proportional-covariance final block. A new full-rank canonical chart and a corrected adapted past remain available.
3. **Objection: equal terminal covariance licenses old-owner reuse. UPHELD against the shortcut.** Projection-nest intertwining, incidence, and trace–bracket matching remain separate requirements.
4. **Objection: the generic line chart supplies the production R-063 forest. UPHELD against that inference.** It has no declared spatial Wick lift or historical low identification.
5. **Objection: future variance and forest can both repair the sign. DISMISSED.** They are alternate coordinates of the same owner and may not be counted twice.

6. **Objection: positive (6.6) proves stability. UPHELD against that inference.** Equation (6.5) is nonzero at small noise, so the origin is not stationary even in the parameter diagnostic; moreover no physical control lift has been specified.
7. **Objection: nonidentifiability proves no complete lift exists. UPHELD against that inference.** It proves that additional owner data are necessary, not mathematical nonexistence.
8. **Objection: the result is phase-specific. DISMISSED.** Every statement concerns chart legality or a signed owner diagnostic. No spatial pattern appears in a theorem premise or conclusion.

11. Result footer

The primary executable derives the registered covariance determinant, canonical prefix ranks, f'' , f''' , $f^{(4)}$, both exact thresholds, the trace-bracket mismatch, the source/sextic relative action, and the orthogonal-gauge nonidentifiability witness. It verifies the Gaussian fixture at three Gauss–Hermite orders.

The non-importing independent executable reconstructs M from the pinned A1 family masses and lock projector, reads the retained coefficients from the pinned R-147 manifest, inverts rational matrices by Fraction Gaussian elimination, uses Jacobi complementary minors, differentiates the row through second-order jet arithmetic and a separate Taylor series, and rejects terminal-prefix, mean-square, denominator, sign, and non-Gaussian mutations. Its matrix and symbolic routes are independent; its numerical mismatch check shares NumPy’s Gauss–Hermite backend with the primary and is not advertised as backend-independent quadrature.

Dependencies: A1, R-125, R-130, R-141, R-142, R-146, R-147.

Exploration ledger: EXP-000625 through EXP-000631. EXP-000630 narrows the rank theorem to the fresh final block, and EXP-000631 separates the coefficient-background parameter Hessian from an unspecified physical source lift.

Negative and audit records:

- NG-2026-08-02-A13-R147-EXACT-CANONICAL-ACTIVE-SPECTATOR-LIFT;
- NG-2026-08-02-A13-ACTIVE-SPECTATOR-JET-OWNER-COMPLETION;
- AUDIT-2026-08-02-A13-R147-ABSOLUTE-DEFECT-AS-RELATIVE-HESSIAN.

No-overclaim statement: R-148 is a fixed-cutoff, fixed-floor fresh-final-block rank obstruction and a separately declared last-root coefficient diagnostic. It proves no rank-one necessity for an adapted past. Its small-noise signs are pointwise at fixed adverse parameters, not uniform at the threshold, and its parameter Hessian is not a physical deterministic-control Hessian. It does not transport the registered multi-root production owners, prove stationarity at the diagnostic origin, establish a global-noise sign, determine the complete owner, close T-050 or A13, prove Nelson, perform removals, construct an interacting measure, select or exclude BCC or any other phase, require PDE replacement, or close Sector A.

Next required action: choose one route before calculating. On the canonical route, specify the full-rank covariance factor, time partition, synthesis, terminal current, output/spatial projectors, covariance ownership, adapted feedback, forest jets, future replicas, balanced atoms, and low owner, and derive every row directly in that chart. On the old-chart route, retain the registered physical prefixes and prove the complete relative residual without claiming canonical transport. In either case add source, sextic, anisotropic endpoint payment, forest/future, and returned low with their registered multiplicities and test the final bound against (8.1).

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_canonical_prefix_rank_active_spectator_lift_relative_hessian_boundary_verify.py
```


Result ID: A13-CLASSII-CANONICAL-PREFIX-RANK-ACTIVE-SPECTATOR-LIFT-RELATIVE-HESSIAN-BOUNDARY; Ledger ID: R-148.

Precise statement: exact canonical-prefix rank obstruction for the R-147 line; exact noncanonical last-root trace-bracket mismatch; exact minimal coefficient-background parameter gradient, Hessian, source, and terminal-sextic diagnostic; physical control Hessian not identified; coefficient-diagonal owner nonidentifiability.

Scope: fixed cutoff and positive floor; registered A1 covariance; separately declared last-root coefficient chart; no old-owner transport.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Expected output: primary and independent PASS; integrated verifier PASS.

Falsification gate: any rank, determinant, Stein factor, derivative, threshold, source/sextic multiplicity, mutation, hash, PDF, or release check fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no complete production owner sign, global-noise theorem, stationarity, T-050, A13, Nelson, removal, interacting measure, phase selection, BCC preference or exclusion, PDE replacement, or Sector-A closure.

Proof complete: false; T-050 closed: false; Sector A closed: false.

Next required action: choose a full-rank canonical reconstruction or the registered old-chart direct route, derive every owner in that one chart, and test the complete signed residual against $\eta < 5/11$ and $\zeta < 13/50$.